

Hands-On lab: Encrypting and Decrypting Email with Mailvelope

Estimated time needed : 40 minutes

Objective(s):

After completing this lab, you will be able to:

- Install Mailvelope
- Generate key pairs
- Create and send an encrypted email
- Decrypt an encrypted email

What you'll need

- A computer running Windows, macOS, or Linux
- Chrome, Firefox, or Edge web browser
- Two different email accounts, one for sending an email and another for receiving the email, each from one of the following providers:
 - Gmail
 - Outlook.com
 - Yahoo
 - Zoho Mail
 - mail.ru

Note: This lab's instructions were created on a Windows computer using Chrome, Gmail, and Outlook.com. However, the steps required will be similar if you use any other software or applications listed above. For technology-specific instructions, including instructions for using email providers other than those listed, see the Mailvelope [FAQ's](#)

Task 1: Install Mailvelope

To use Mailvelope, you must first install the Mailvelope browser extension.

1. Access <https://mailvelope.com/>
2. Click Download Mailvelope.



Communicating securely with Mailvelope



Mailvelope is a **browser add-on** that you can use in **Chrome**, **Edge** and **Firefox** to securely encrypt your emails with PGP using **webmail providers**

**Download
Mailvelope**

Note: If downloading the package is challenging or if you prefer to proceed exclusively with the Firefox browser, please [Click Here](#).

3. Click **Add to Chrome** or the comparable button for the browser that you are using.

[Home](#) > [Extensions](#) > Mailvelope



Mailvelope



www.mailvelope.com



Featured

★★★★★ 430 ⓘ

Social & Communication

100,000+ users

4. In the Add **Mailvelope** pop-up window, click **Add extension**.



Add "Mailvelope"?

It can:

Read and change all your data on all websites

Add extension

Cancel

5. Click the **Extensions** icon on your browser toolbar, and then select **Mailvelope** from the list of extensions.



6. A pop-up window opens, which prompts you to configure Mailvelope. Click Let's start!



Task 2: Generate key pairs

Now you'll generate two key pairs. One is for the sender account, the email account that you will use to send the encrypted email. The other is for the receiver account, the email account that you will use to receive and decrypt that email.

1. Click Generate key.



2. Enter your name and email address for the sender account, and then create a password. (Write down the password or store it in a password manager; you will need it to complete the lab.) Finally, click Generate.

Make Mailvelope even more secure by personalizing your security background.

[Personalize now](#)[< Key Management](#)

Generate Key

Name

Full name of the key owner

Email

[Advanced >>](#)

Enter Password

Re-enter Password

☐ Upload public key to Mailvelope Key Server (can be deleted at any time). [Learn more](#)

[Go](#)

Note that the password you create is for your private key. Mailvelope does not store your password, so you cannot recover it if you lose it! Instead, you must delete the generated key pair and create a new one.

3. Now you'll generate a key pair for the receiver account. On the Key Management page, click **Generate**.

The screenshot shows the Mailvelope Key Management page. At the top, the Mailvelope logo is on the left, and navigation links for 'Key Management' (highlighted), 'Encrypt', 'Decrypt', and 'Options' are on the right. Below the navigation bar, a grey banner contains the text 'Make Mailvelope even more secure by personalizing your security background.' and a red 'Personalize' button. The main content area has a purple background with a repeating pattern of server racks and geometric shapes. A white card in the center is titled 'Key Management'. Inside the card, there are five buttons: '+ Generate', 'Import', 'Search', 'Export', and 'Refresh'. Below these buttons is a table with three columns: 'Name', 'Email', and 'Key ID'. The table contains one entry with a key icon, a blurred name, a 'Default' status button, an email address ending in '@gmail.com', and a blurred key ID. At the bottom left of the page, the copyright notice '© 2012–2022 Mailvelope GmbH' is visible.

Mailvelope

Key Management Encrypt Decrypt Options

Make Mailvelope even more secure by personalizing your security background. **Personalize**

Key Management

+ Generate Import Search Export Refresh

Name	Email	Key ID
 [blurred] Default	[blurred]@gmail.com	[blurred]

© 2012–2022 Mailvelope GmbH

4. Enter your name and email address for the receiver account, and then create a new password. Finally, click **Generate**.

Make Mailvelope even more secure by personalizing your security background.

[Personalize now](#)[< Key Management](#)

Generate Key

Name

Full name of the key owner

Email

[Advanced >>](#)

Enter Password

Re-enter Password

☐ Upload public key to Mailvelope Key Server (can be deleted at any time). [Learn more](#)

[Go](#)

5. Now you should have two keys listed on the Key Management page. Your sender account should be labeled Default.

Make Mailvelope even more secure by personalizing your security background.

[Personalize now](#)

Key Management

+ Generate

Import

Search

Export

Refresh

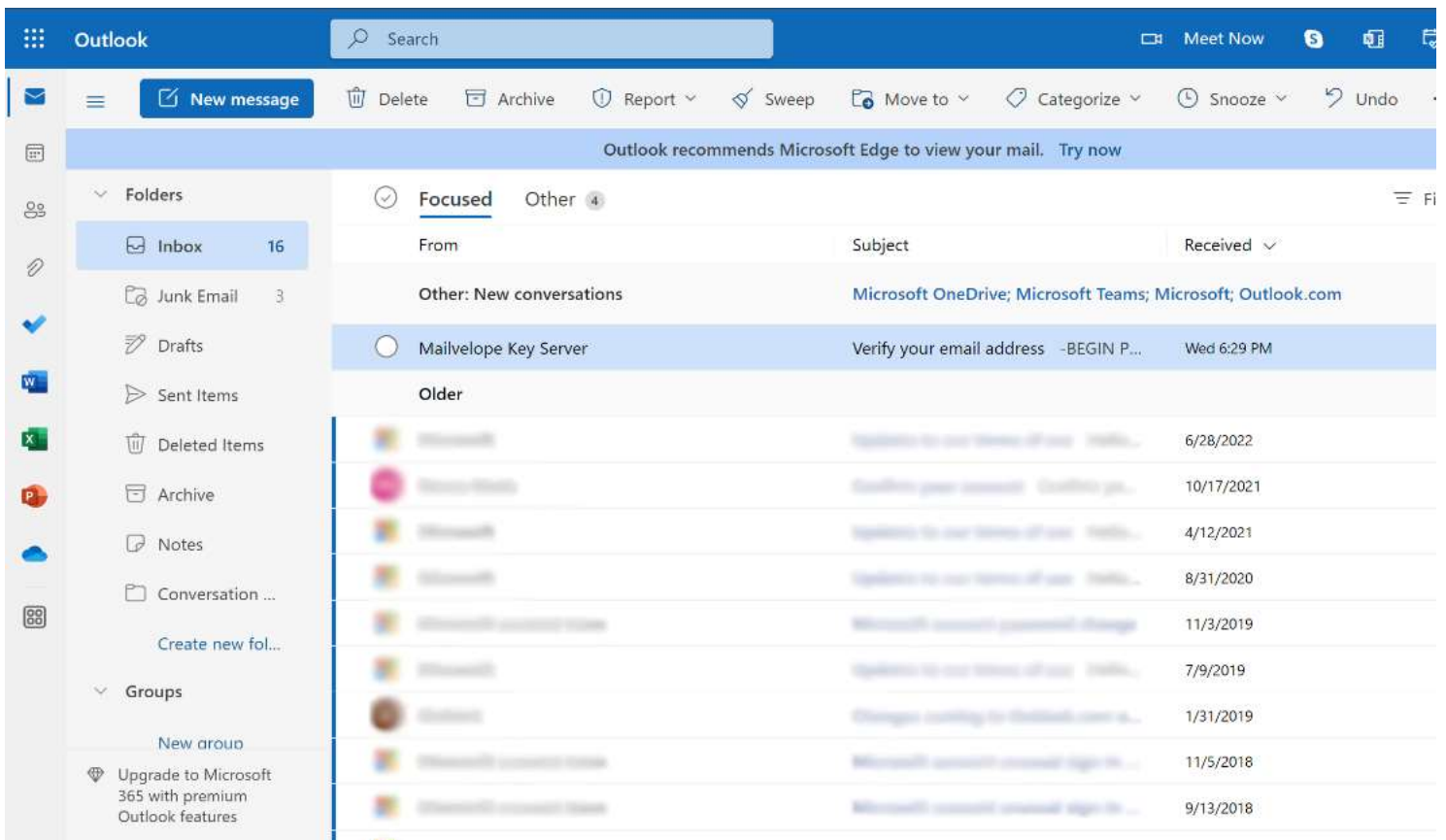
Filters: All

Name	Email	Key ID	Created	
 Default	@gmail.com		2022-11-02	>
	@outlook.com		2022-11-02	>

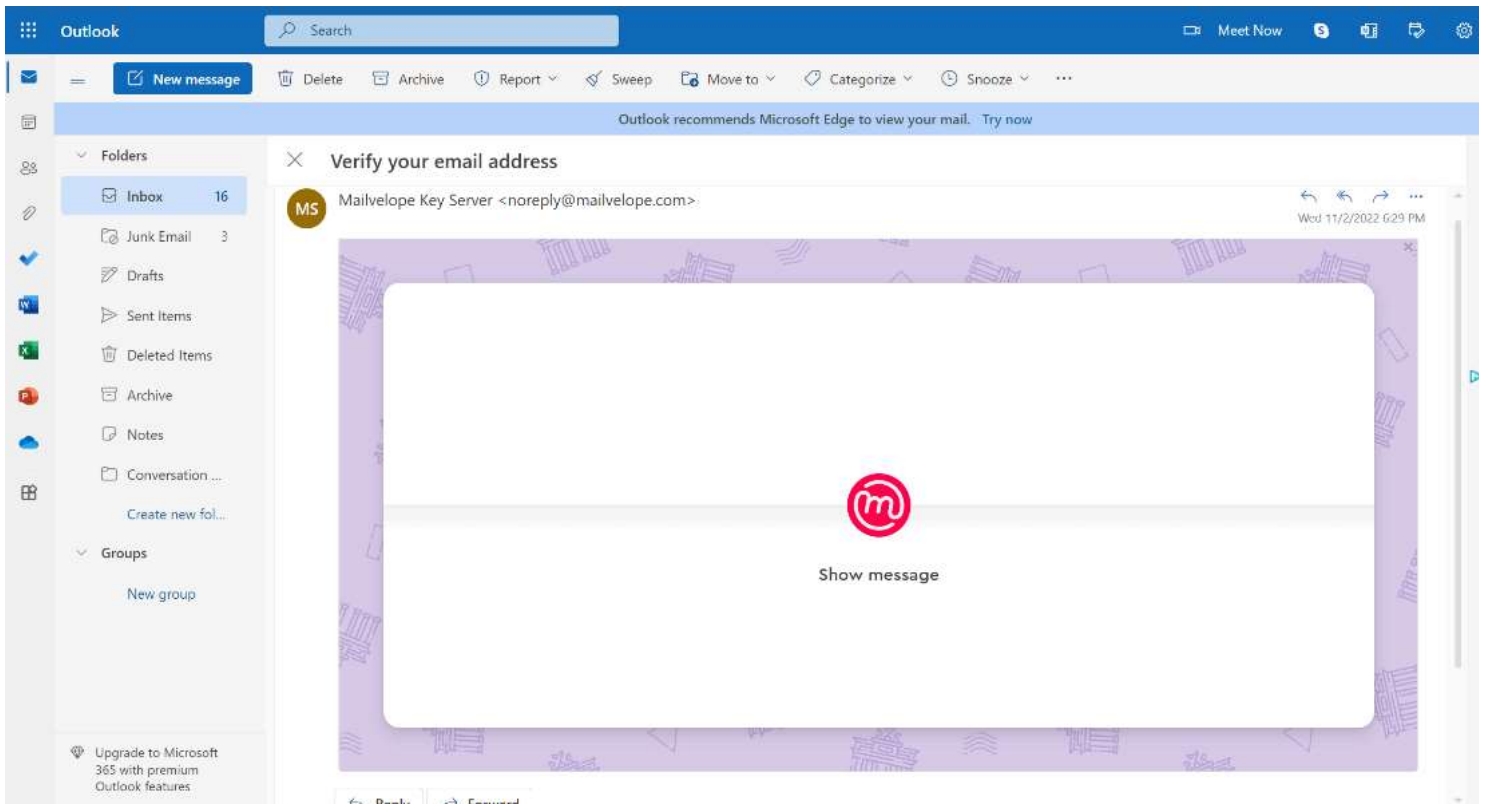
Task 3: Verify your email addresses

After generating your key pairs, you need to verify your email address for each pair.

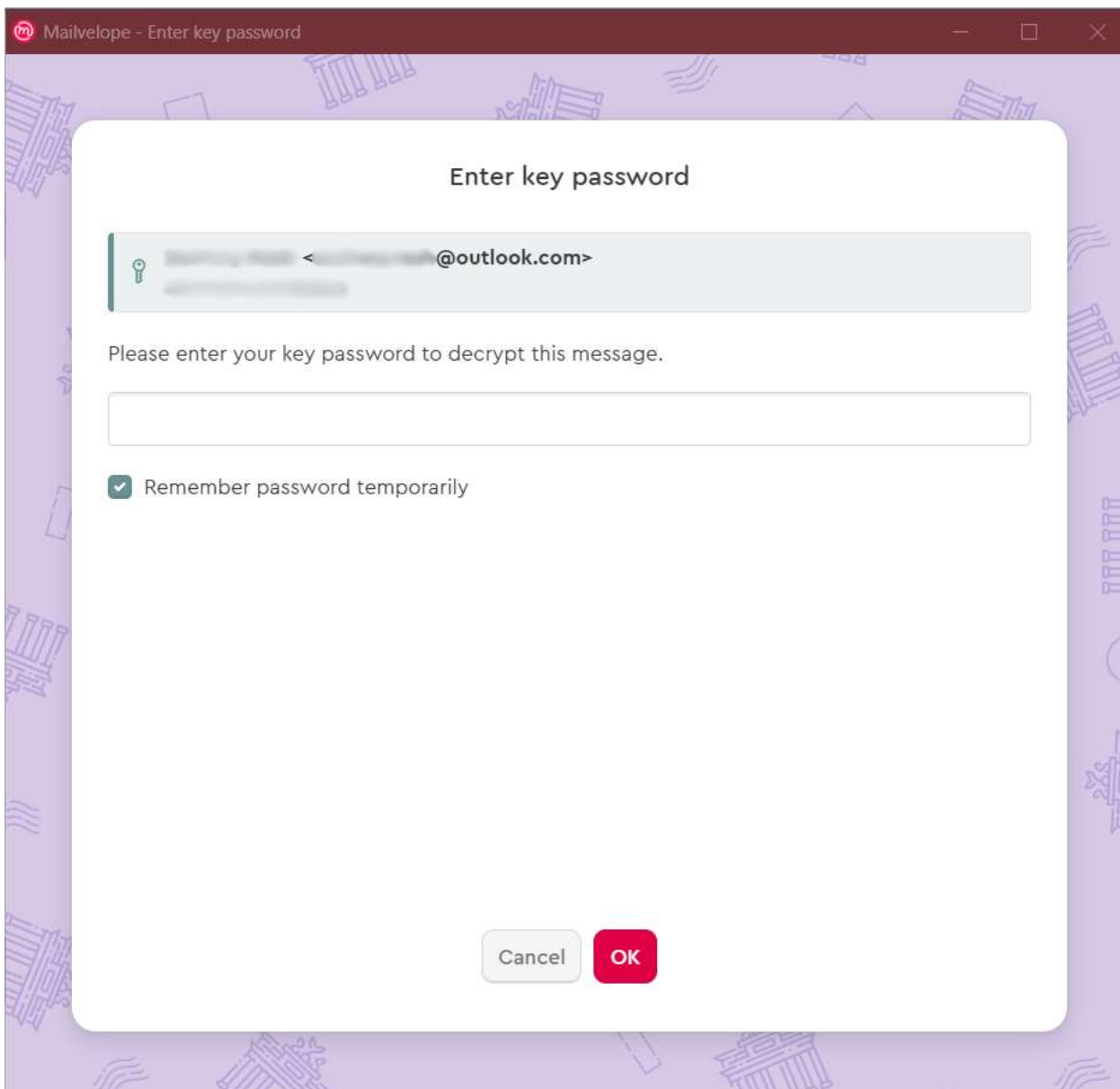
1. Use your browser to log into the sender account, which is an outlook.com account in this demonstration.
2. On the Inbox page, select the email from Mailvelope Key Server with the subject **“Verify your email address”**.



3. Click **Show message**, and then click the provided link to verify your email address.



4. In the “Enter key password” pop-up window, enter the password you used when generating your Mailvelope key pair for this email account, and then click OK.



5. A window appears that states that you successfully verified this email account. Now complete Task 3 for the receiver account.

Mailvelope Key Server

Email address [redacted]@outlook.com successfully verified

Your public OpenPGP key is now available at the following link: [https://keys.mailvelope.com/pks/lookup?op=get&search=\[redacted\]@outlook.com](https://keys.mailvelope.com/pks/lookup?op=get&search=[redacted]@outlook.com)

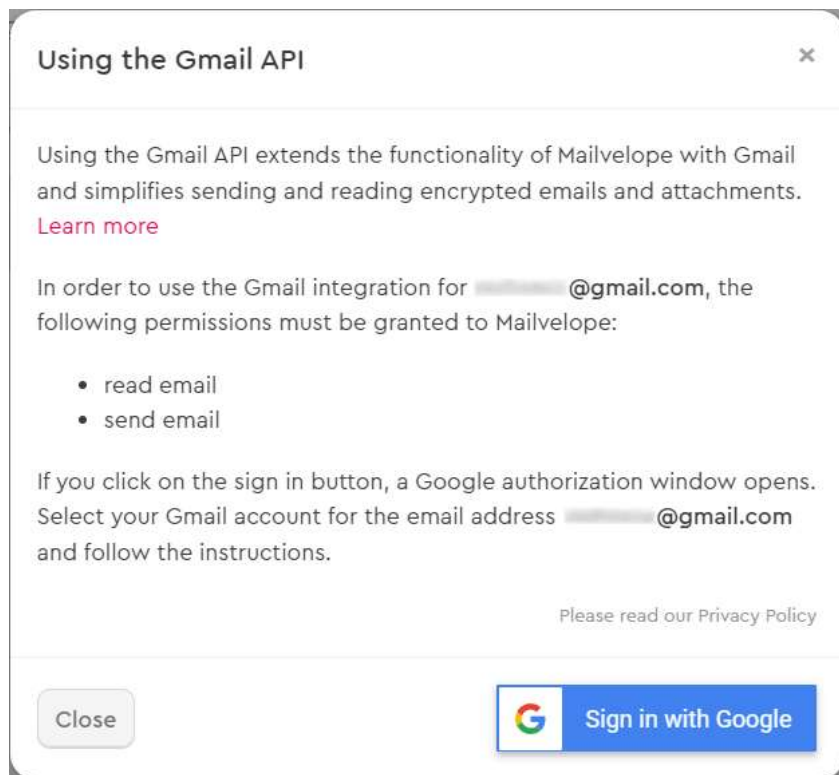
Optional: Enable the Gmail API for Mailvelope

If you are using Gmail for one of the accounts, you must complete a few extra steps before using it with Mailvelope.

1. Log into your Gmail account.
2. Click the Mailvelope icon, which is next to Compose.



3. In the Using the Gmail API pop-up window that opens, click **Sign in with Google**

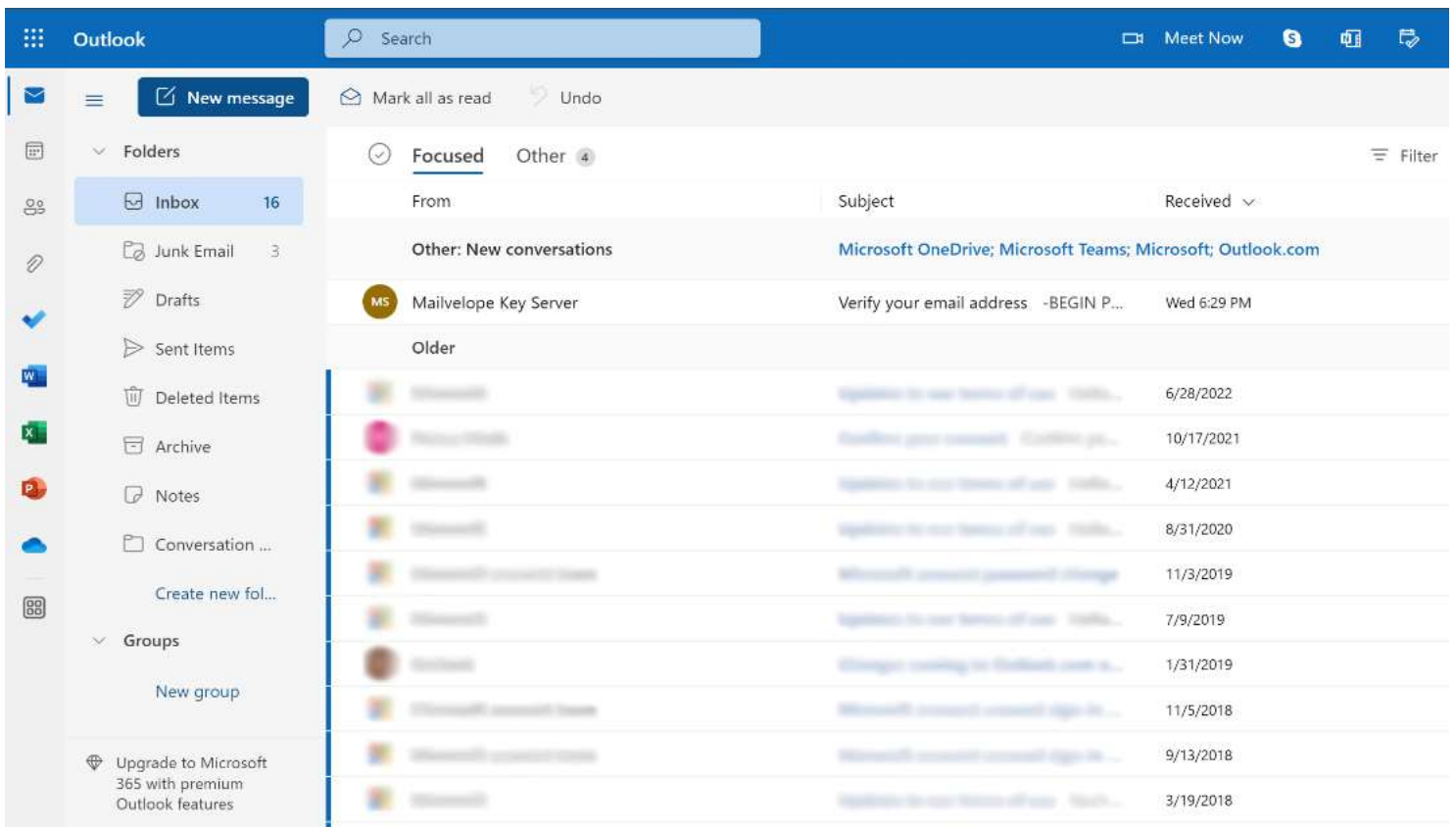


4. Follow the on-screen prompts to log into your Google account and grant Mailvelope access.

Task 4: Create and send an encrypted email

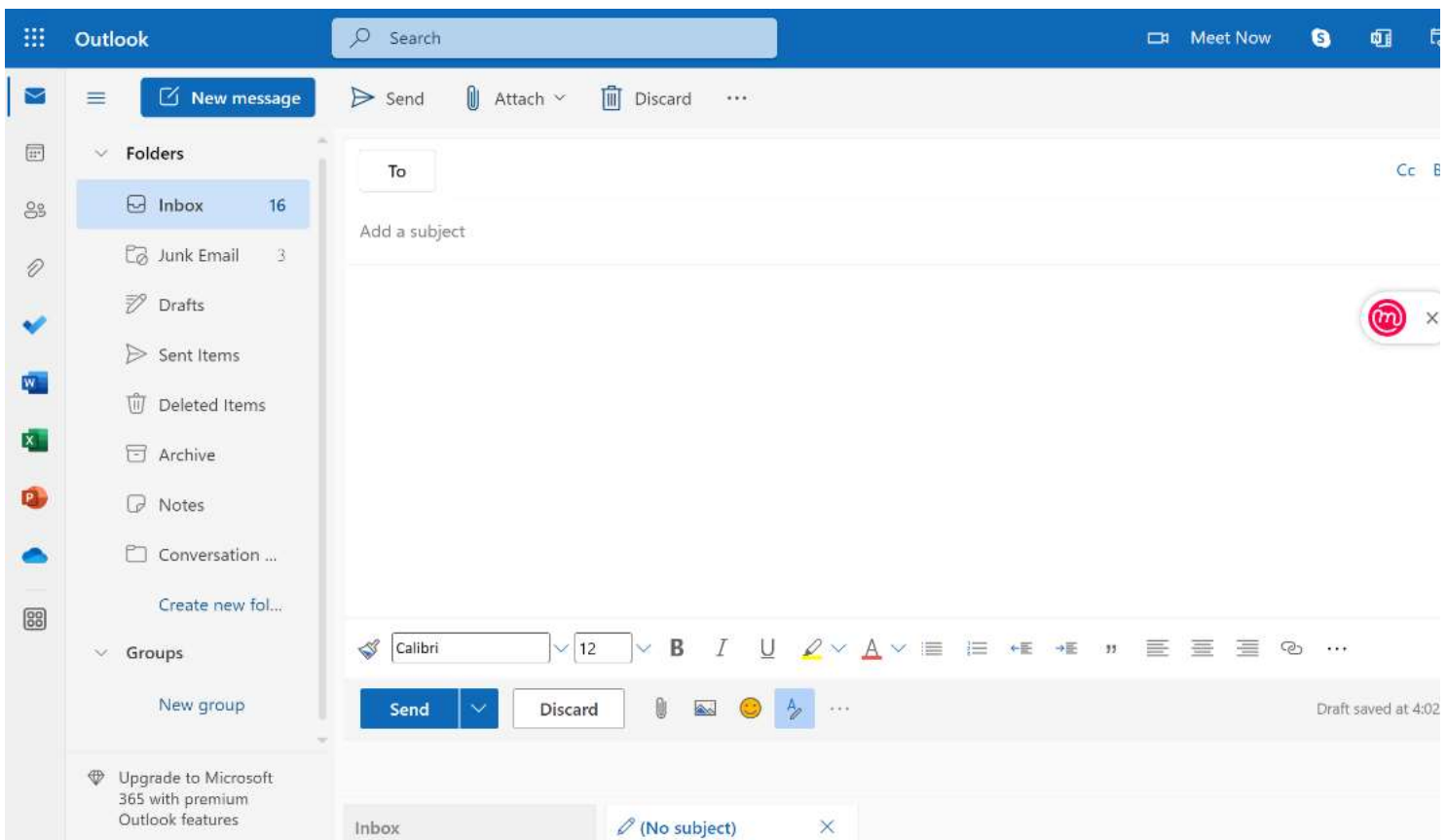
Now you'll use the sender account to create and send an encrypted email.

1. Use your browser to log into the sender account, which is an outlook.com account in this demonstration.
2. To compose an encrypted message first, click **New message** or the comparable button on the Inbox page of your email account.



Note that in Gmail and some other email accounts, you can skip step 2.

3. If you have successfully set up Mailvelope for this email account, a Mailvelope icon is displayed.



4. Click the Mailvelope icon to start writing a secure email.

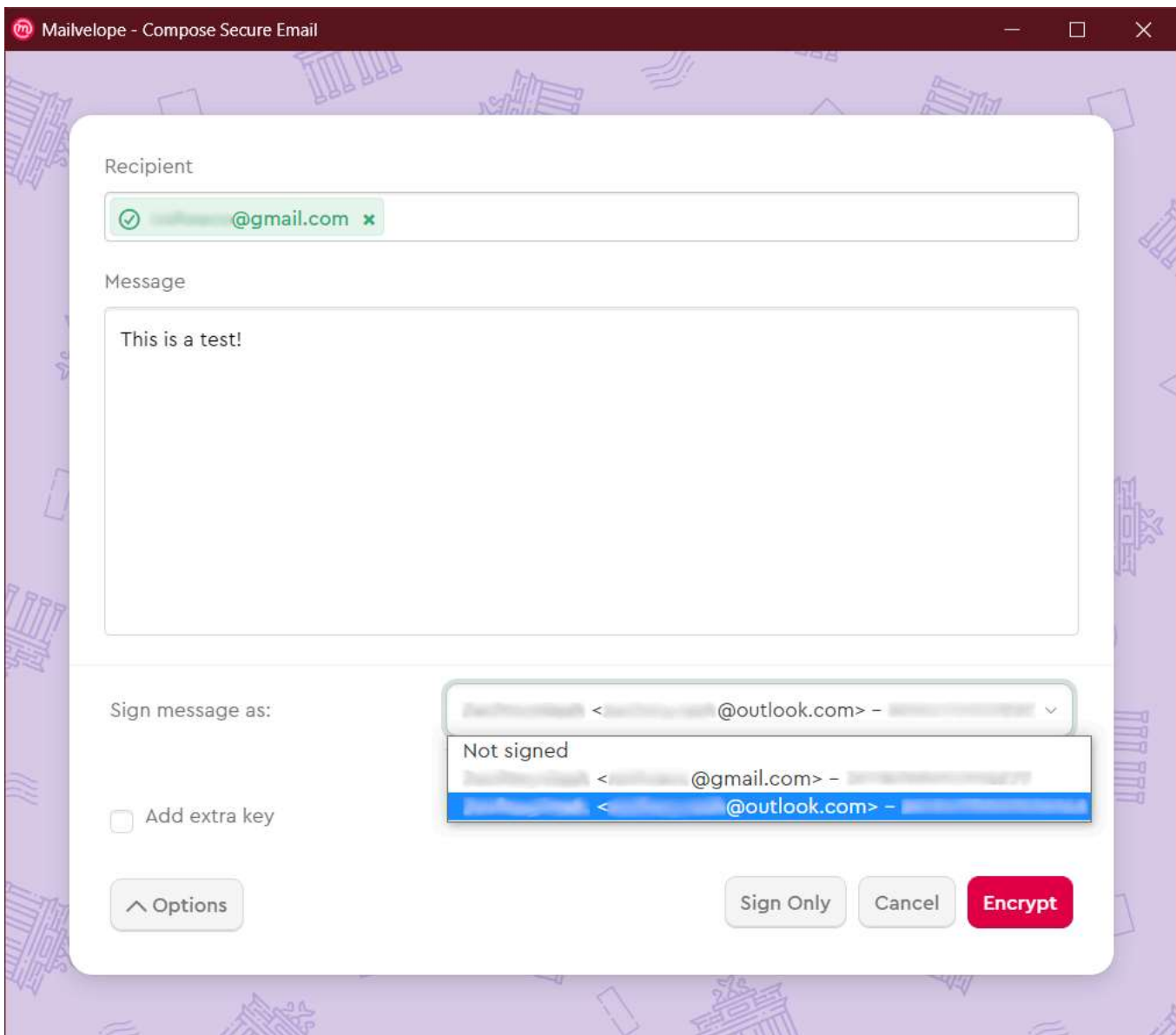


5. In the Compose Secure Email pop-up window, enter the receiver email address and a test message.

The image shows a window titled "Mailvelope - Compose Secure Email". The window has a purple background with a repeating pattern of white icons representing various digital security concepts like keys, locks, and documents. In the center is a white rounded rectangle representing the email composition area. At the top of this area is a label "Recipient" above a text input field containing "@gmail.com". Below the input field is a preview of the email header: "Subject: [redacted] <[redacted]@gmail.com> - [redacted]". Below the header is a large text area containing the message "This is a test!". At the bottom of the white area are three buttons: "Options" with a downward arrow, "Sign Only", and "Encrypt" in a red button.

Note that Mailvelope automatically searches for the receiver's public key as you enter their email address. As long as you know the email address associated with the receiver's public key, you can send them encrypted messages.

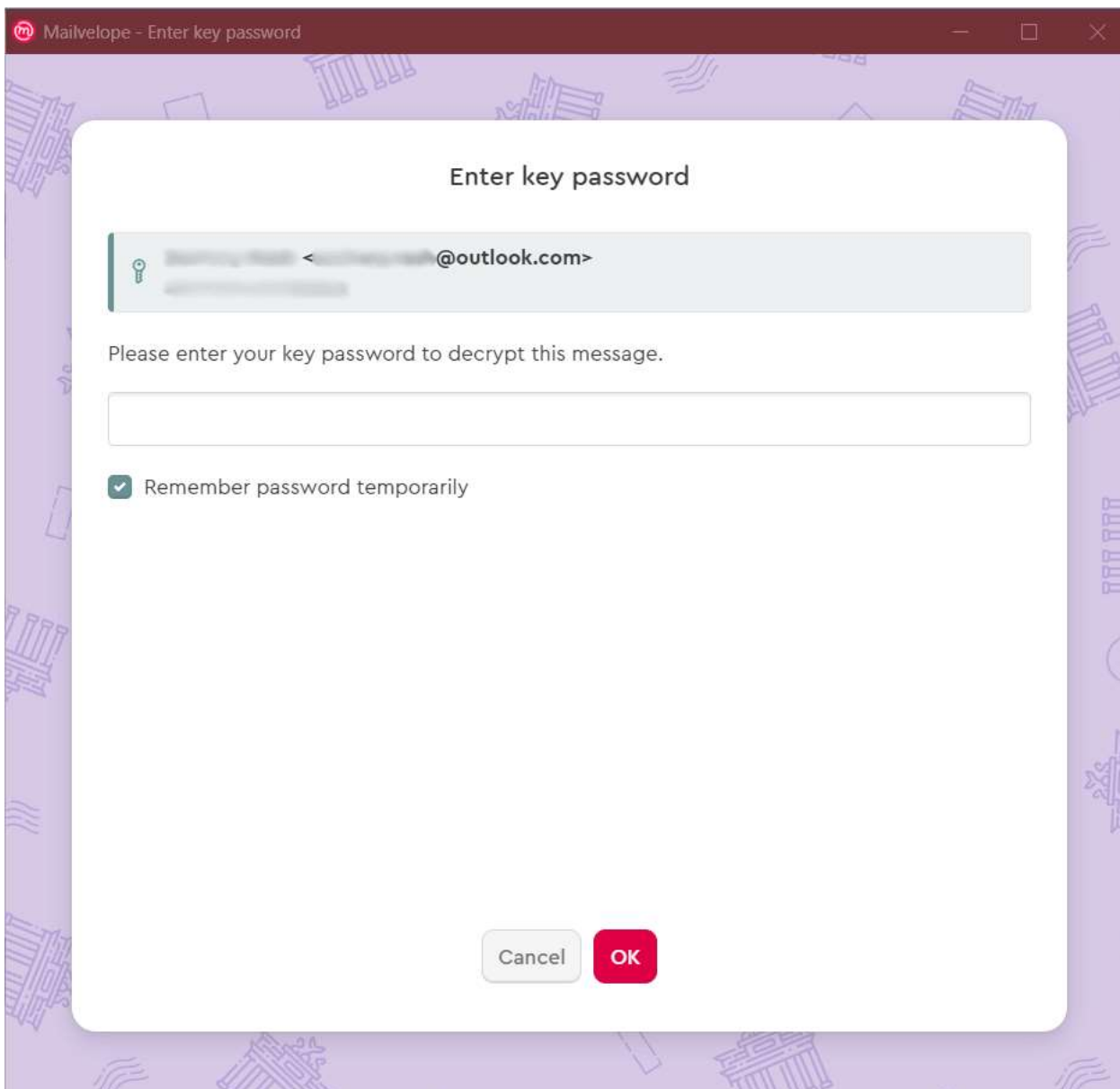
6. Confirm that you are signing this message with the sender account's digital signature. Click Options, and then select your sender account from the **Sign message** as list.



Note that your default digital signature is the signature for the account listed as default on the Key Management page.

7. Click **Encrypt**.

8. In the “Enter key password” pop-up window, enter the password for the sender account, and then **click OK**.



9. Encrypted text is displayed now in the email body.

To

Test

-----BEGIN PGP MESSAGE-----
Version: Mailvelope v4.6.1
Comment: <https://www.mailvelope.com>

wcFMA2zQUSnNYunNAQ//cQ/eq+b1MC6+OVdAABtvS01Re7UrpDLUZo5HVd10
W36UKsPWeVrHiWIsGsJhU16kGE99iUFny/2KwvJYedr0lsC6JZjoKiI+eHP0
ESSd4v88mkrIhwY0uitiHFSx3zm1Xrod60xbtX1N47sppsDKE6Bw3JgA7apw

Calibri 12 **B** *I* U ...

10. Enter the receiver email address in the **To** field and a subject in the **Subject** field. Next, click **Send**.

To

██████████@gmail.com

Test

Calibri

▼

12

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Send

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Discard

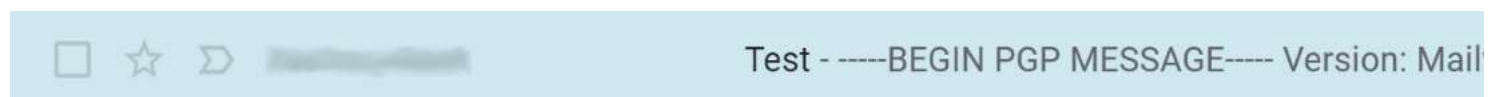
A

...

Task 5: Decrypt an encrypted email

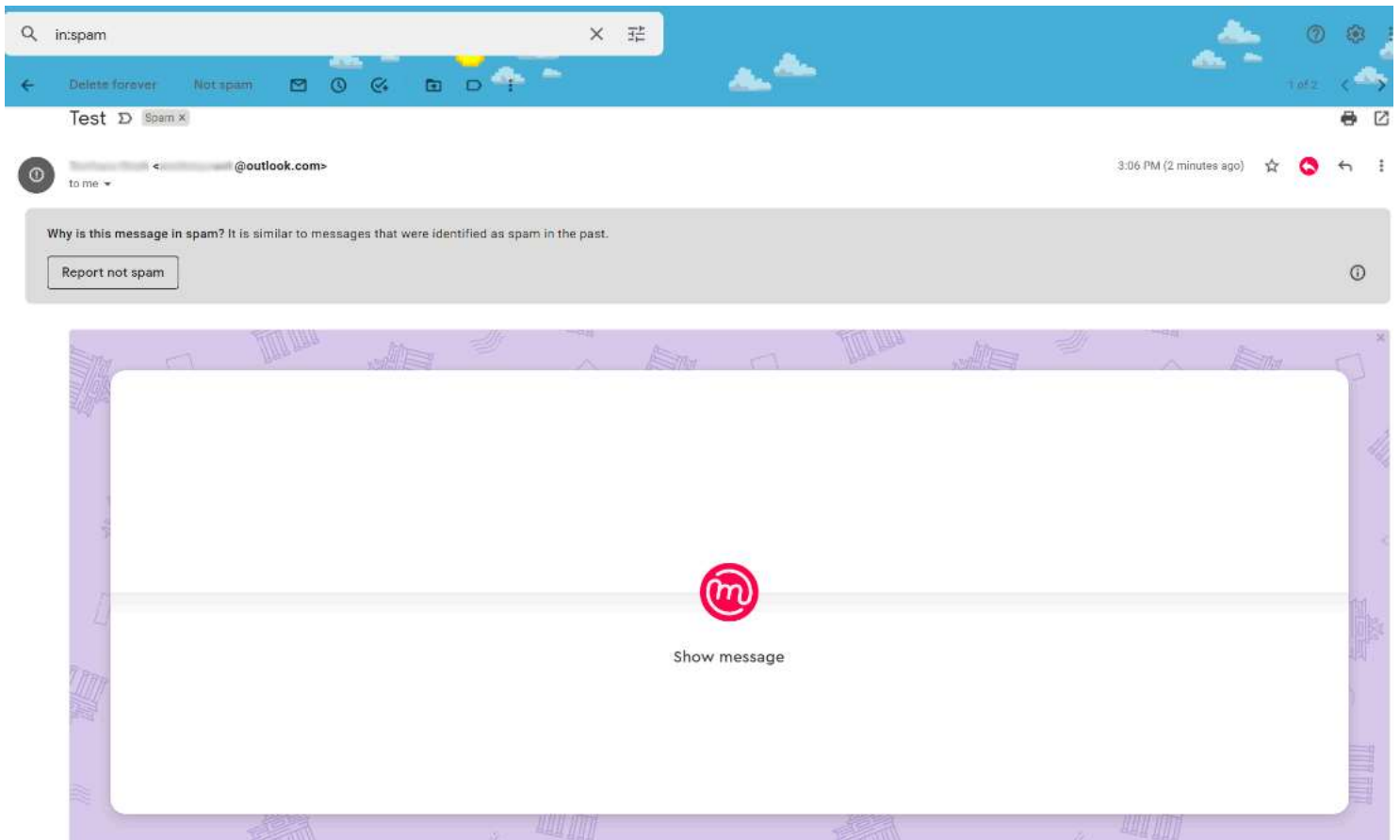
Now you'll use the receiver account to open and decrypt the email.

1. Log into the receiver account, which is a Gmail account in this demonstration.
2. Locate and select the encrypted email from the sender account.

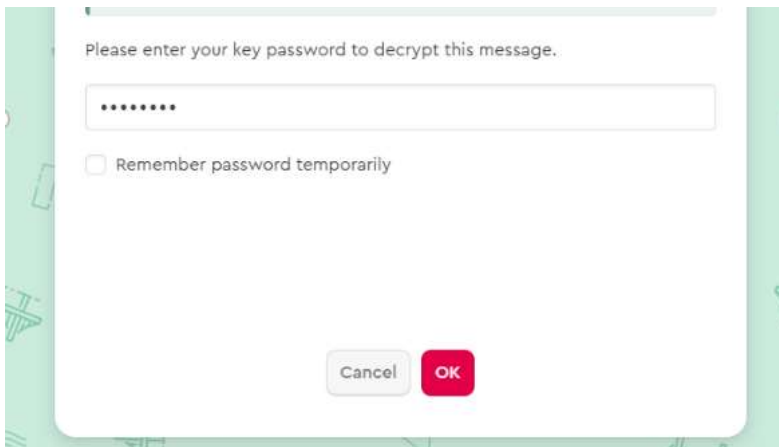


Note that the encrypted email might be in your Spam or Junk folder.

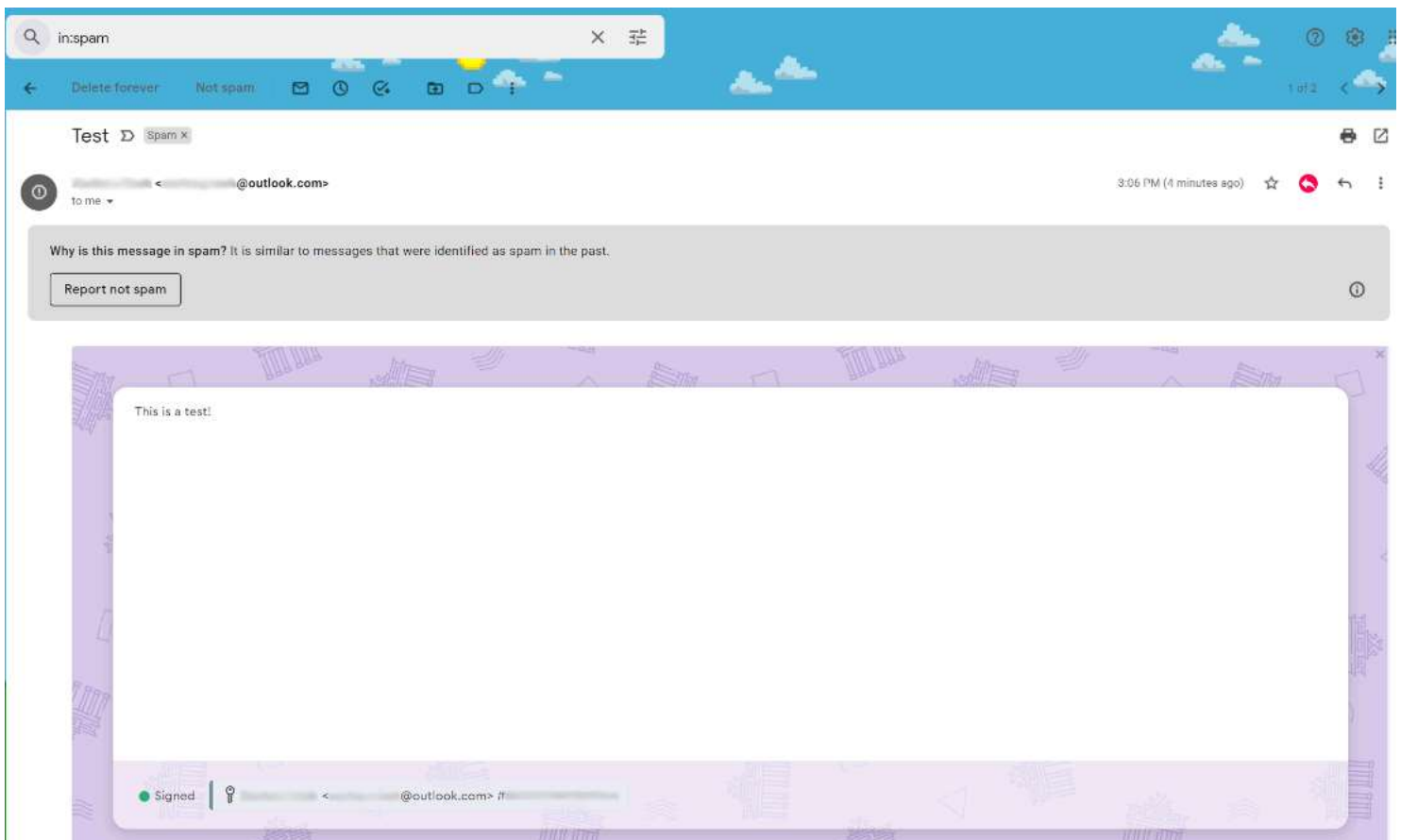
3. Click **Show message**.



4. Enter your key password to decrypt the message, and then **click OK**.



5. Your decrypted message is displayed, and the sender's digital signature is confirmed.



6. Optional: If you have a second browser or an email app, use it to open the same email. Without the Mailvelope browser extension, the message is encrypted, and you do not have the option to decrypt it.

Conclusion

Congratulations! You just used public key encryption to send, receive, and read an encrypted message. To do so, you installed Mailvelope and used it to generate key pairs for email accounts. Next, you used Mailvelope and those key pairs to create and send an encrypted email and decrypt the email from a receiver account.

You can now use Mailvelope to encrypt messages between you, your friends, your coworkers, and anyone else interested in secure communication. To send someone an encrypted message, enter the person's email address in the Compose Secure Email pop-up window: Mailvelope will automatically locate their public key. Alternatively, you can import their public key on the Key Management page. Access the Mailvelope [website](#) for more about importing public keys and other Mailvelope features.

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Changelog

Date	Version	Changed by	Change Description
2023-11-29	1.0	Sowmyaa Gurusamy	Created initial version

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